

Experiment 01

NMOS Characterization

Analog IC Design Lab — Cadence Virtuoso (GPDk090)

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1. Objective

To characterize the DC behaviour of an NMOS transistor (gpdk090_nmos1v, $W = 1\mu\text{m}$, $L = 100\text{nm}$) in Cadence Virtuoso by observing:

- Output characteristics — drain current (I_d) versus drain-source voltage (V_{ds}) for a family of gate-source voltages (V_{gs}).
- Transfer characteristics — drain current (I_d) versus gate-source voltage (V_{gs}) at a fixed operating point.
- The transition of the device between the triode (linear) and saturation regions of operation.

2. Schematic

The test bench consists of a single NMOS transistor with independent DC voltage sources V1 (driving V_{gs} at the gate) and V0 (driving V_{ds} at the drain), with source and bulk tied to ground.

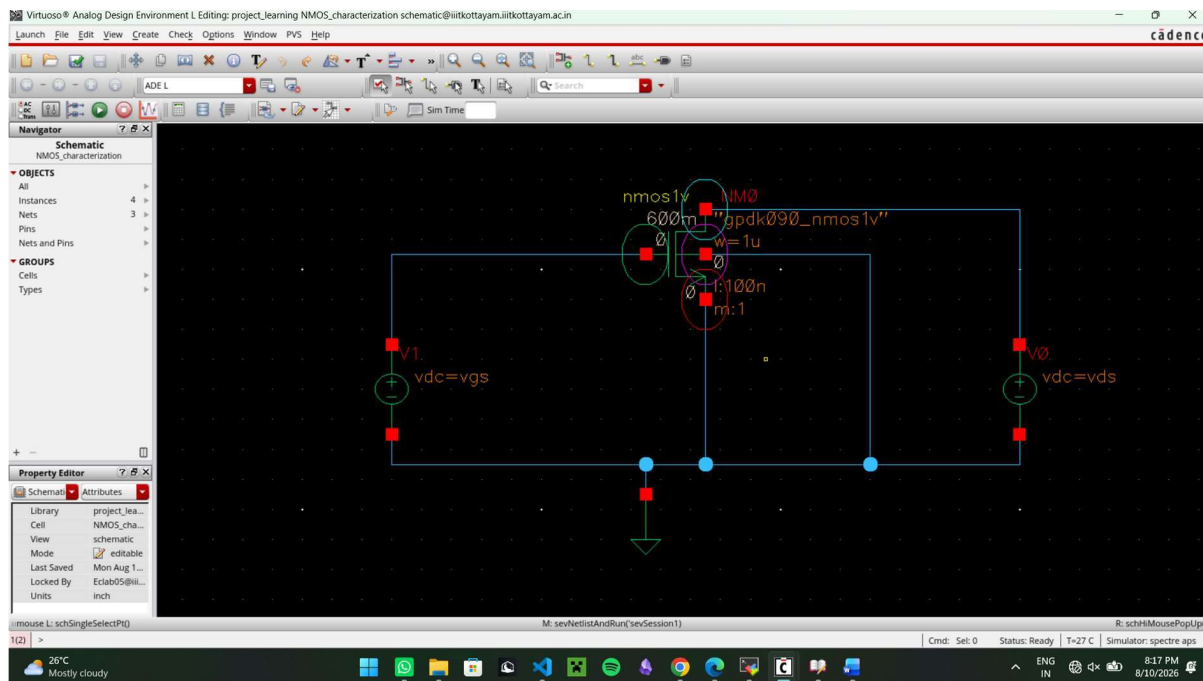


Figure 1. NMOS characterization schematic — nmos1v (gpdk090_nmos1v), $W = 1\mu\text{m}$, $L = 100\text{nm}$, $m = 1$.

Parameter	Value
Device	gpdk090_nmos1v
Width (W)	$1\mu\text{m}$
Length (L)	100nm
Multiplier (m)	1
Gate driver	V1, vdc = vgs (swept)

Drain driver	V0, vdc = vds (swept)
Source / Bulk	Grounded

3. Simulation Setup

Simulations were run in ADE L using DC analysis with automatic start-stop sweep of V_{ds} from 0 to 1.2 V. V_{gs} was swept as a family variable to generate the output characteristic curves, and separately held for the transfer characteristic sweep. Node currents and voltages at all four terminals (S, G, B, D) of NM0 were saved and plotted.

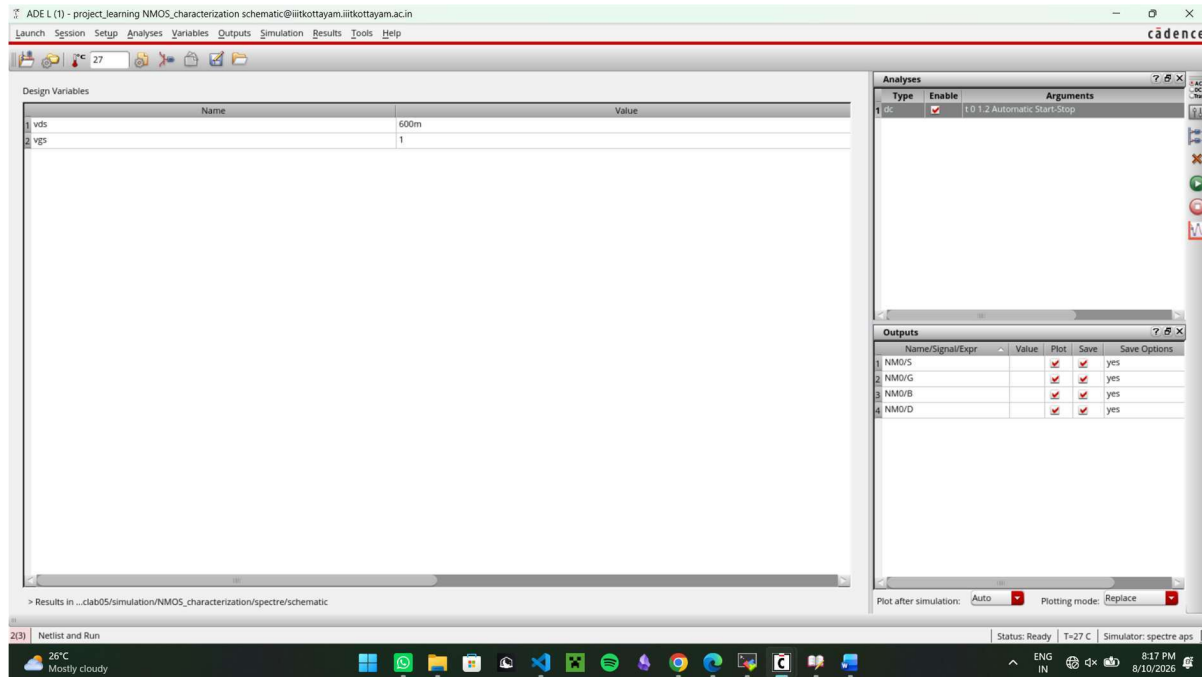


Figure 2. ADE L setup — design variables (v_{ds} , v_{gs}) and DC analysis configuration.

4. Results

4.1 Output Characteristics (I_d vs V_{ds} , family of V_{gs})

V_{ds} was swept from 0 V to 1.2 V, with V_{gs} stepped as a family parameter. Each trace corresponds to a fixed V_{gs} .



Figure 3. I_d - V_{ds} output characteristics for a family of V_{gs} values.

Observation: each curve rises steeply at low V_{ds} (triode region, I_d roughly proportional to V_{ds}) and then flattens out once V_{ds} exceeds $V_{gs} - V_{th}$ (saturation region, I_d becomes largely independent of V_{ds}). Higher V_{gs} traces saturate at a higher current level, consistent with expected square-law/short-channel NMOS behaviour.

4.2 Transfer Characteristics (I_d vs V_{gs})

V_{gs} was swept from 0 V to 1.2 V at a fixed V_{ds} .



Figure 4. I_d - V_{gs} transfer characteristic.

Observation: I_d stays close to zero until V_{gs} crosses a threshold (visually around 0.25–0.3 V on this plot), after which current rises with increasing slope, consistent with the device turning on and moving toward strong inversion. The curve was not linearized to formally extract V_{th} — see Limitations below.

5. Analysis

- The output family confirms the two expected regions of operation: triode (V_{ds} small, resistive-like rise) and saturation (V_{ds} large, current levels off).
- The transfer curve confirms qualitative turn-on behaviour but a numeric threshold voltage was not extracted using the linear-extrapolation or g_m/I_d method.
- No g_m or g_m/I_d vs V_{gs} plot was generated — this would normally follow directly from the swept I_d - V_{gs} data by differentiating in ADE L's calculator (`dd(...)` function).

6. Limitations / Not Yet Completed

In the interest of an accurate submission, the following were not completed in this pass:

- Formal V_{th} extraction (linear extrapolation of the max-slope tangent to the V_{gs} axis, or g_m/I_d method).
- g_m and g_m/I_d vs V_{gs} (or vs I_d) plots — relevant since the group op-amp project uses g_m/I_d sizing methodology.
- Log-scale I_d - V_{gs} plot to inspect the subthreshold slope.
- A written interpretation connecting these curves back to hand-calculated square-law expectations.

7. Conclusion

The DC output and transfer characteristics of the `gpd090_nmos1v` device were simulated successfully in Cadence Virtuoso, and the triode-to-saturation transition and gate turn-on behaviour were observed as expected. Threshold voltage extraction and g_m/g_m-I_d analysis remain to be added to complete the experiment.